Chapter 3 - Rings and Moduls of Fractions

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Exercise 1

(i) ($\Rightarrow$): Let M be generated by $\{m_1, m_2, \dots, m_n\}$, for $\forall i \in [1, n] \cap \mathbb{Z}$, we have

$$\forall r, t \in S, \exists s \in S, \text{s.t. } m_i s(r - t) = 0, \text{i.e. } m_i s = 0$$

Denote the s above as s_i. Since M is finitely-generated, we can let

$$s = s_1 s_2 \dots s_n$$

And have $sM = 0$.

(ii) ($\Leftarrow$): When such s exist, then for $\forall \frac{m_1}{s_1}, \frac{m_2}{s_2} \in S^{-1}M$,

$$s(s_1 m_2 - s_1 m_2) = 0 \implies \frac{m_1}{s_1} = \frac{m_2}{s_1}$$

So $S^{-1}M = 0$.

$\square$

Exercise 2

By *Proposition 1.9*, the problem is equivalent to prove that for any $\frac{a}{s} \in S^{-1}A$, we have

$$\forall \frac{b}{t} \in S^{-1}A, 1 - \frac{a}{s}\frac{b}{t} \text{is a unit in } S^{-1}A$$

Since $1 + \mathfrak{a}$ is a multiplicative set, $\exists c \in \mathfrak{a}, \text{s.t. } st = 1 + c$. Now

$$\begin{aligned} 1 - \frac{a}{s}\frac{b}{t} &= 1 - \frac{ab}{st} \\ &= 1 - \frac{ab}{1+c} \\ &= \frac{1 + c - ab}{1 + c} \end{aligned}$$

And we can easily see $\frac{1+c}{1+c-ab} = 1 + \frac{ab}{1+c-ab} \in S^{-1}A$. So the inverse of $1 - \frac{a}{s}\frac{b}{t}$ is $1 + \frac{ab}{1+c-ab}$. $\square$

Exercise 3

Define

$$f : (ST)^{-1}A \longrightarrow U^{-1}(S^{-1}A)$$

by mapping

$$\frac{a}{st} \longmapsto \frac{\frac{a}{s}}{\frac{t}{1}}$$

We'll prove f is an isomorphism:

(i) Surjection: Trivial.

(ii) Injection:

$$\frac{\frac{a_1}{s_1}}{\frac{t_1}{1}} = \frac{\frac{a_2}{s_2}}{\frac{t_2}{1}}$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } \frac{t}{1}\left(\frac{a_1 t_2}{s_1} - \frac{a_2 t_1}{s_2}\right) = 0$$

$\Leftrightarrow$

$$\exists t \in T, \text{s.t. } t(a_1 t_2 s_2 - a_2 t_1 s_1) = 0$$

$\Leftrightarrow$

$$\frac{a_1}{s_1 t_1} = \frac{a_2}{s_2 t_2}$$

So f is injective.

(iii) Homomorphism: Trivial.

$\square$

Exercise 4

Define

$$f : S^{-1}B \longrightarrow T^{-1}B$$

by

$$\frac{b}{s} \longmapsto \frac{b}{f(s)}$$

We'll prove it's an isomorphism:

(i) Homomorphism: Trivial

(ii) Surjection: Trivial

(iii) Injection:

$$\frac{b_1}{f(s_1)} = \frac{b_2}{f(s_2)}$$

$\Leftrightarrow$

$$\exists s \in S, \text{s.t. } f(s)\left(b_1 f(s_2) - b_2 f(s_1)\right) = 0$$

$\Leftrightarrow$

$$\frac{b_1}{s_1} = \frac{b_2}{s_2}$$

$\square$

Exercise 5

(i) By *Proposition 3.11-(v)*, the operation S^{-1} commutes with the formations of radicals. So $\forall \mathfrak{p} \subseteq A$,

$$(\mathrm{Nil}(A))_{\mathfrak{p}} = \mathrm{Nil}(A_{\mathfrak{p}}) = 0$$

Therefore by *Proposition 3.8*, $\mathrm{Nil}(A_{\mathfrak{p}}) = 0$.

(ii) No. Counterexample given here

Exercise 6

(i) Σ is nonempty, therefore has maximal elements by *Zorn's Lemma*.

(ii) First, we need to show $A \backslash S$ is an ideal.